X-UAD: A Unified, Uncertainty-Aware and Self-Explaining Detector for Physical and Digital Face Attacks

***Abstract*—Face recognition systems can be defeated by presentation attacks staged in front of the camera, such as a printed photograph, a replayed video or a silicone mask, and by digital manipulation of the captured image through face swapping, expression reenactment, synthesis or adversarial perturbation. A detector built for one family is usually blind to the other, and the recent unified detectors that handle both still share three practical weaknesses. They emit a single uncalibrated score with no way of signalling that an input resembles nothing they were trained on; they cannot point to *where* in the frame the evidence sits; and when they do explain themselves, the explanation comes from a language model that never observed the detector's internal state. We present X-UAD, one model aimed at all three. Adaptive Quantum Attention adds a Born-rule scoring channel beside ordinary dot-product attention. It is sign-invariant by construction, which matches the interference character of moiré, halftone and checkerboard evidence, and it costs $O(dL)$ parameters rather than $O(d^2)$. The same learnable rotations define an orthonormal evidence basis whose measurements serve simultaneously as classification features and, completely unchanged, as the explanation shown to a reader. A decoupled evidential head predicts class direction and evidence magnitude separately, and pseudo-unseen evidence suppression trains that magnitude on blends of two attack families, an outlier proxy built from the attack taxonomy itself. On a difficulty-calibrated, identity-disjoint benchmark of eight attack families with pixel-level ground truth, X-UAD reaches 1.46% ACER against 0.71% for the strongest baseline, while giving the best calibration among the working detectors (1.94% ECE) and the best evidence localisation (77.8% region-level IoU). On attack families withheld from training it achieves the highest average detection rate (77.5%), and the synthetic-blend objective alone is worth 5.4 AUROC points on the uncertainty channel. We also report, without softening it, that every method in the comparison collapses under a serious change of capture domain. Code, the benchmark generator and configuration files mapping the same pipeline onto public capture-based datasets are released.**

***Index Terms*—face anti-spoofing, presentation attack detection, deepfake detection, unified attack detection, vision transformer, quantum-inspired attention, evidential deep learning, uncertainty estimation, open-set recognition, explainable AI**

I. Introduction

Consider everything that can go wrong in front of a face-unlock camera. Someone could hold up a printed photo of you. They could replay a video of your face on a phone screen. They could wear a silicone mask shaped like your face. These are *physical* attacks, and each leaves a physical fingerprint behind: the halftone dot pattern of a printer, the pixel grid of a phone screen aliasing into a moiré pattern, the waxy sheen and flattened texture of silicone.

Now consider what can go wrong *after* the camera has captured a real, unmanipulated photograph. Someone could swap in a different identity. They could drive your expression with someone else's performance. They could generate a face that never belonged to any real person. They could add a barely visible pattern of noise that flips a classifier's decision without a human noticing anything at all. These are *digital* attacks, and their fingerprints look entirely different: blending seams where a swapped face meets the original, resolution mismatches, checkerboard patterns left behind by the upsampling layers of a generator, unusual peaks in the frequency spectrum.

Because these two sets of fingerprints look nothing alike, the two research communities that study them grew up separately. Face anti-spoofing work focused on texture and depth cues [1–4]. Face forgery research focused on frequency artifacts and blending cues [5–8]. The split is understandable, but it is not something you can deploy. An operator needs one decision, one threshold and one audit trail, not two models that might disagree with no principled way to combine them. A recent line of work [9–13] has finally built unified corpora and joint detectors that handle both attack families at once, which is real progress. It also exposed a problem that had been hiding underneath. On the attack types these models were trained on they are essentially saturated, with ACER around 0.5 to 1.4%, but the moment you hold out an attack family during training and show it only at test time, error rates jump to between 22 and 44% with large variance between runs [9]. Unified detection, in other words, is solved in-distribution and still wide open out-of-distribution.

Layered on top of that gap, we see three capabilities missing

from almost all of this literature, and all three are things an operator genuinely needs the moment a score comes back ambiguous rather than obviously fake or obviously real.

The first is a real notion of not knowing. A softmax output is fundamentally a closed-set quantity, since it has to redistribute all of its probability mass across the classes it was trained on. A completely novel kind of attack therefore gets assigned high confidence toward whichever known class it happens to resemble most. What is actually needed is a second-order belief, one that is allowed to sit close to empty when the input does not look like anything the model has learned.

The second is evidence localisation. Depth-map supervision [1,3] works reasonably for physical attacks, since a printed photo or a replayed video really is flatter than a real face. But depth is a proxy for liveness rather than a map of forensic evidence, and it says nothing useful about a digitally swapped face that came from a perfectly live, perfectly three-dimensional original capture.

The third is an explanation that is actually tied to the decision. Recent work has started feeding face images into multimodal language models and asking them to describe what looks fake [14, 15]. The results read fluently, but the language model never sees the detector's internal state. It is reasoning over the raw image, independently of whatever the classifier actually computed. Its narrative is therefore correlated with the decision at best, and it is free to mention artifacts such as "a visible screen bezel" that are not actually present in the frame or that the detector never used.

A. What this paper does

We built X-UAD, a single model that detects physical and digital face attacks together, reports how much it does not know, points to where the evidence is, and explains its own decision in language that traces back to numbers it actually computed. Three ideas make this work.

Adaptive Quantum Attention (AQA). Ordinary attention scores a pair of tokens with an unbounded linear functional, the dot product. A great deal of the evidence for face attacks, though, is fundamentally an interference phenomenon: a moiré beat, a checkerboard pattern, a spectral peak. These show up in the *squared magnitude* of a projection, not in the raw projection, and their sign carries no information. A moiré fringe and its photographic negative are the same evidence, but a dot product treats them as opposites. AQA adds a second scoring channel based on the Born rule from quantum measurement. Each token is represented as a unit vector, a "pure state", evolved by a small stack of learnable rotation matrices (a Givens circuit), and two tokens are compared by the squared overlap between their evolved states. This channel is mixed into the ordinary channel through a learned, per-head, input-conditioned gate. The rotation matrices are exactly orthogonal by construction and are parameterised by $O(dL)$ angles rather than the $O(d^2)$ weights of a normal linear layer.

A quantum evidence basis that makes the explanation part of the computation. A learnable orthonormal set of basis vectors, built the same way as the image of the standard basis under a rotation circuit, stays exactly orthonormal at every step of training and is measured against every spatial token. The resulting probabilities form a token-by-mode evidence map that is used as a classification feature *and*, completely unchanged, as the explanation. The explanation is therefore a direct read-out of the computation rather than a post-hoc approximation of it produced by a separate model.

Pseudo-Unseen Evidence Suppression (PUES). If you take two training images from two different attack families and blend them, you get something no real camera or generator could ever produce, because no physical process makes a superposition of a print attack and a face swap. That makes these blends a free, built-in proxy for "genuinely novel input". We train the model to push its evidential output toward zero on these blends, which teaches the uncertainty channel what "I do not recognise this" should look like without ever showing the model a real unseen attack.

We also built **evidence-grounded reasoning**. Every forward pass emits a typed, numeric evidence record, and the natural-language explanation is generated only from that record, either by a deterministic rule-based generator whose hallucination rate is zero by construction, since every sentence is gated by a numeric condition, or optionally by a hosted multimodal language model whose output is then automatically checked against the record.

Finally, because no existing capture-based dataset marks *where* in the frame the attack evidence lives, we built a difficulty-calibrated, identity-disjoint benchmark of eight attack families with pixel-level evidence masks, along with identity-consistent, leave-one-attack-out and cross-domain protocols, and an explicit calibration procedure so the benchmark cannot be trivially solved.

Section II surveys the research threads X-UAD tries to bring together and states the gap explicitly. Section III describes the benchmark. Section IV describes the model. Section V reports the experiments. Section VI returns to the gap and explains, piece by piece, what closes it.

II. Related Work and the Gap

A. Physical presentation attack detection

Early face anti-spoofing treated liveness as a texture classification problem. Auxiliary supervision [1] replaced the plain real/fake label with a pseudo-depth map and an rPPG signal as extra training targets. Central Difference Convolution [2] modified the very first layers of the network to respond to illumination-invariant micro-texture by mixing in the local gradient response, and it remains one of the strongest lightweight backbones for physical attacks specifically. Pixel-wise supervision was revisited more systematically in [3]. Several works then tackled the harder problem of generalising across capture domains, meaning different cameras and different lighting, using adversarial domain generalisation [16], single-side domain generalisation [17], and style-shuffled feature separation [4]. All of this work shares one assumption: that the attack happens in front of the camera. None of it has any mechanism for a capture that is physically

genuine but was manipulated afterward.

B. Digital forgery detection

FaceForensics++ [5] and Celeb-DF [6] are the benchmarks this line of work grew up around. The most transferable cue turned out to be spectral. Generative models leave periodic artifacts from their upsampling layers [18], which show up as distinctive peaks in the Fourier magnitude [7], and this is a large part of why fairly simple detectors generalise surprisingly well across different generator architectures [8]. The symmetric weakness applies here too. These detectors have no concept of a re-acquisition medium and fail badly on printed or replayed attacks.

C. Unified physical-digital detection

UniAttackData [9] is arguably the most important recent contribution in this space. It is an identity-consistent corpus in which the same person appears both genuine and under every attack type, so that identity itself cannot become a shortcut, paired with a vision-language detector using teacher-student prompting. The CVPR'24 and ICCV'25 challenges built on this [10, 11], scaling to nearly 2,900 subjects and 54 attack types, and confirmed two things worth remembering: methods built on large pretrained multimodal backbones dominate the leaderboard, and single-modality CNNs get stuck above 20% ACER on the harder protocol. UniDefense [13] approaches the same goal through dual-space reconstruction. What none of these systems has is an uncertainty channel, an evidence map, or a genuine explanation. All that comes out is a score.

D. Uncertainty and open-set recognition

Maximum softmax probability [19] is a weak out-of-distribution signal for exactly the reason described above, since softmax is a closed-set construction. Prior networks [20] and evidential deep learning [21], built on subjective logic [22], place a Dirichlet distribution over the simplex so that total evidence, and therefore vacuity, becomes something a model can actually output. Outlier exposure [23] showed that having even a rough proxy for "something else" during training is worth more than any purely intrinsic uncertainty score. Open-set single-domain generalisation has recently been explored for anti-spoofing on its own [24], but nobody had combined evidential uncertainty with unified physical-digital detection before this work, and no prior work builds its outlier proxy directly out of the attack taxonomy the way PUES does.

E. Explainability and MLLM-based forensics

Attention rollout [25] and deletion/insertion faithfulness testing [26] are the standard tools for checking whether an explanation actually reflects a model's decision. FFAA [14] fine-tunes a multimodal LLM for open-world forgery analysis with a multi-answer decision system, and M2F2-Det [15] bridges CLIP and an LLM for interpretable forged-face detection. Both produce fluent text, and both are, in a real sense, narrating after the fact. The language model sees the image but never the detector's internal evidence, so there is no way to check its account against the decision it is supposedly explaining, and it is free to describe artifacts that simply are not there.

F. Quantum-inspired attention

Quantum Vision Transformers [27] and quantum self-attention [28, 29] replace ordinary linear projections with parameterised quantum circuits, reporting accuracy comparable to classical vision transformers at a small fraction of the parameters, which is essentially an $O(n^2) \to O(n)$ argument. This line of work is motivated primarily by parameter efficiency and is evaluated on generic image classification. It has never been connected to a task whose evidence is literally an interference pattern, and none of it offers an explainability story.

G. Stating the gap

Every one of the six lines above occupies a proper subset of what an operator actually needs, and the missing pieces are not simply additive. Unifying the attack space creates the uncertainty problem, because more families means more ways to fall outside all of them. Adding uncertainty creates the explanation problem, because an operator asked to adjudicate a vacuous, low-confidence sample needs to know what the model actually saw. And an explanation is only worth trusting if it is tied to the same internal evidence that produced the decision in the first place. X-UAD is designed around this dependency chain rather than as a grab-bag union of separate modules bolted together.

III. The X-UAD Benchmark

A. Why we built our own benchmark

The capture-based corpora the unified-detection literature relies on [9, 11] are access-restricted, and more importantly for our purposes, none of them marks where in the frame the attack evidence actually sits. Training and scoring an evidence localiser needs ground truth that a real capture pipeline simply cannot provide, because nobody knows exactly which pixels a real printer or a real generator changed. We therefore build a benchmark where the evidence is injected by a known, documented operator, so its support is known exactly, by construction.

We want to be upfront about the trade-off this involves. A generated benchmark buys pixel-level supervision, perfect identity consistency, and full reproducibility, and it costs external validity, since an operator is a *model* of a physical or generative process rather than the process itself. We try to manage that risk in four ways. Every operator is motivated by the actual physics or generator architecture it stands in for. The benchmark is difficulty-calibrated so it cannot be solved trivially. We keep a second, genuinely different capture domain purely for cross-domain testing. And we release configuration files that map the identical taxonomy, protocols and model code onto CASIA-FASD [30], OULU-NPU [31], SiW-Mv2, CelebA-Spoof [32], FaceForensics++ [5], Celeb-DF-v2 [6] and UniAttackData [9], so that every claim in this paper is directly re-testable against real captured data by anyone who obtains access to those corpora. Nothing about the model, the losses, the training loop, the

explainability code or the reasoning module is specific to our synthetic benchmark.

B. Source faces and identity discipline

Domain A (LFW-Sim) is built from 12,794 identity-labelled, tightly cropped face photographs of 5,616 subjects from Labeled Faces in the Wild [33]. Splits are assigned by a stable hash of the subject identity, so train, validation and test never share a subject. Every source face contributes exactly one genuine sample and one sample of each of the eight attack families, so the mutual information between identity and label is zero by construction and no detector can cheat by memorising who a person is rather than learning what an attack looks like. The source imagery is grayscale, and we treat that as a deliberate constraint rather than a limitation. It removes colour-based shortcuts such as display gamut shift, ink gamut compression and colour-transfer residue, and forces every method, ours included, to make its decision on texture, geometry and frequency evidence instead.

Domain B (Wild-Sim) applies the identical operators to 700 in-the-wild colour face crops at low and variable resolution with an unknown prior JPEG compression history. It is never used for training. Its only job is to be the target of the cross-domain protocol, and it differs from Domain A in colour, alignment, resolution and compression history all at once.

C. Attack taxonomy

Eight classes span the two super-classes. Each operator takes a global strength parameter $s \in (0,1]$ that scales every artifact's amplitude, and returns a soft mask $M \in [0,1]^{H\times W}$ marking which pixels carry the injected evidence.

- **print** (physical): Yule-Nielsen tone transfer with dot gain, an ordered halftone screen at a sampled line frequency and angle, paper substrate grain, loss of live micro-texture, a small planar perspective shift, and a simulated re-acquisition chain comprising signal-dependent shot and read noise followed by JPEG compression.
- **replay** (physical): display gamut and contrast compression, an RGB sub-pixel stripe lattice at a sampled period and tilt that aliases into a visible low-frequency moiré beat, rolling-shutter banding, specular backlight glare, then re-acquisition and double compression.
- **mask3d** (physical): edge-preserving smoothing confined to the skin region, since silicone loses live micro-texture, with eye and mouth cut-outs through which the wearer's real features remain visible, a waxy albedo with broad specular sheen, and a seam at the mask's rim.
- **faceswap** (digital): a donor face aligned to the target's face box, passed through a deliberately low generator resolution and upsampled to mimic the resolution mismatch of a real swap pipeline, colour-transferred, and alpha-blended into the inner-face region. The residual evidence is the blending boundary and resolution mismatch, not gross misalignment.
- **reenact** (digital): a smooth non-rigid warp confined to the mouth and eye regions, with a local sharpness and contrast inconsistency inside the manipulated area only.
- **synth** (digital): the classic generator signature, namely over-smoothed skin combined with over-sharpened detail, a transposed-convolution checkerboard at a sampled stride [18], and an explicit azimuthally modulated radial peak injected into the Fourier magnitude [7].
- **adversarial** (digital): not random noise, but a genuine ℓ_∞ PGD perturbation ($\varepsilon = 6/255$, 7 steps) [34] crafted against an independently trained surrogate detector, a small VGG-style network that shares no weights, architecture or random seed with any model we evaluate, and is fit only on the training split. The perturbations seen at validation and test time are therefore a genuine transfer, that is, black-box, attack. The surrogate itself is flipped to "genuine" on 38.8% of the crafted samples. Ground truth for localisation is the smoothed, normalised magnitude of the perturbation itself.

D. Difficulty calibration

A simulated benchmark is worthless if a simple linear probe can solve it outright. We fix each operator's strength through a single documented procedure: sweep s and pick the largest value at which a reference detector (CDCN++, our strongest lightweight texture baseline) does *not* saturate, leaving measurable headroom on both the binary and the eight-way task. This lands on $s = 0.62$, where the reference detector reaches 1.92% ACER but only 91.0% eight-way accuracy and 3.74% expected calibration error, with a worst-case per-family APCER of 2.67%. In other words the binary problem sits close to, but not at, the ceiling, while family discrimination, calibration and unseen-attack behaviour all remain wide open. This mirrors, on a benchmark we fully control, the exact pattern reported for UniAttackData [9]: a saturated seen-attack protocol next to a wide-open unseen-attack one.

E. Protocols

P1 (seen attacks). All eight families appear in train, validation and test, with identity-disjoint splits. This measures closed-set unified detection and family discrimination.

P2 (leave-one-attack-out). One family is withheld entirely during training and appears only at test time. We report ACER on the full test set, the detection rate of the held-out family at the fixed operating threshold, and the AUROC with which the model's own uncertainty channel separates the held-out family from genuine samples, which is to say whether the model knows that it does not know.

P3 (cross-domain). Train on Domain A, test unchanged on Domain B, with zero adaptation.

All thresholds are fixed at the equal-error point of the validation split and applied unchanged to test. Every metric follows ISO/IEC 30107-3 [35]: APCER is the worst case across attack families, BPCER is the genuine-rejection rate, and ACER is their average.

IV. Method

A. Overview

X-UAD maps a face crop $x \in \mathbb{R}^{3\times H\times W}$ to a tuple

$$x \longmapsto \big(\alpha,\, u,\, M,\, \Pi,\, r,\, \tau\big),$$

where $\alpha \in \mathbb{R}^{K}_{>0}$ are Dirichlet concentrations over the $K = 8$ classes, $u \in [0,1]$ is the vacuity (a single scalar answering "how much do I have to go on"), M is the spoof-evidence map, Π is the quantum evidence spectrum over spatial tokens and basis modes, r is the risk score, and τ is the natural-language rationale. All six come out of a single forward pass. A central-difference stem and a spectral branch produce tokens; a stack of Adaptive Quantum Attention blocks processes them; the quantum evidence basis is measured against the output tokens; an evidential head, a localisation decoder and a risk calibrator read everything out; and the reasoning module turns the record into text.

B. Central-difference and spectral evidence stem

Central Difference Convolution [2] interpolates between an ordinary convolution and its local gradient response,

$$y(p_0) \;=\; \sum_{p\in\mathcal{N}} w(p)\,x(p_0+p) \;-\; \theta\Big(\sum_{p\in\mathcal{N}} w(p)\Big)x(p_0), \qquad (1)$$

which amplifies the intensity-invariant part of the response. Note that the extra term is just a single 1×1 convolution, not a second full convolution, so it is nearly free. Four strided CDC stages take a 112^2 input down to 7^2, exposing the intermediate 28^2 and 14^2 maps to the localisation decoder later on.

Periodic structure, meaning the display lattice behind a replay attack or the upsampling checkerboard behind a generated face, is not represented well in a 7×7 spatial token grid; it gets aliased away. We add a compact spectral branch. The log-magnitude DFT of the luminance channel is pooled onto a polar grid (16 radii $\times$ 12 angles), which is translation-invariant and rotation-equivariant by construction, and a small two-layer MLP encodes this profile into four spectral tokens that get appended to the spatial tokens. The same pooled profile is also exported into the evidence record, so the radial peak frequency, its prominence, and the azimuthal anisotropy become concrete, quotable numbers in the eventual explanation.

C. Adaptive Quantum Attention

Amplitude encoding. For a head with dimension d, queries and keys are ℓ_2-normalised to unit vectors, $|\psi_i^Q\rangle = q_i/\|q_i\|$ and $|\psi_j^K\rangle = k_j/\|k_j\|$, a real-valued amplitude encoding of $d = 2^n$ amplitudes across n notional qubits.

Parameterised rotation circuit. A depth-L circuit acts on a ring of amplitudes. Layer ℓ rotates disjoint adjacent pairs, alternating which pairs it touches between layers, each rotation governed by a single learnable angle:

$$G(\vartheta) = \begin{pmatrix} \cos\vartheta & -\sin\vartheta \\ \sin\vartheta & \cos\vartheta \end{pmatrix}, \qquad U(\boldsymbol{\vartheta}) = \prod_{\ell=1}^{L} \bigotimes_{k} G(\vartheta_{\ell k}). \quad (2)$$

This is the real-amplitude analogue of a hardware-efficient ansatz built from nearest-neighbour two-qubit gates with periodic boundary conditions. Alternating the two offsets keeps the circuit connected, so the set of reachable rotations grows toward the full orthogonal group as L increases. Three things matter about this construction. First, it is exactly unitary, since U is a product of rotations, so it stays orthogonal to machine precision at every step of training. We measured $\big|\,\|Ux\| - \|x\|\,\big|_\infty < 10^{-6}$ throughout, so no re-orthogonalisation or penalty term is ever needed. Second, it is parameter-efficient, costing $L\lfloor d/2\rfloor$ angles, that is $O(dL)$, rather than the $O(d^2)$ of a dense projection matrix, the same asymptotic argument made for quantum self-attention elsewhere [27, 29]. Third, because it exactly preserves the vector's norm, the resulting similarity score below is a genuine probability rather than just a bounded number.

The Born channel. Queries and keys are evolved by two *different* circuits. This matters, because if they shared one circuit the rotation would cancel out of the inner product entirely and add nothing. The similarity score is the Born-rule transition probability,

$$S_{ij} \;=\; \big|\langle\psi_i^Q|U_Q^\top U_K|\psi_j^K\rangle\big|^2 \;\in\; [0,1]. \qquad (3)$$

This is a quadratic similarity. It is invariant to a sign flip of either state, and it is maximised equally by alignment *or* anti-alignment. That is exactly the invariance an interference pattern has. A moiré fringe and its negative carry the same forensic information, and it is precisely the invariance an ordinary dot product lacks.

Adaptive gating. Not every kind of evidence is interference-like. A blending boundary, for instance, is really just a plain edge. A per-head gate, conditioned on simple statistics of the token set, decides the mixture between the two channels:

$$g = \sigma\big(W_g[\mathrm{mean}_t(X)\,\|\,\mathrm{std}_t(X)] + b_g\big) \in [0,1]^{H_{\text{heads}}}, \qquad (4)$$

$$A = \mathrm{softmax}\Big((1-g_h)\tfrac{QK^\top}{\sqrt{d}} + g_h\beta S\Big), \qquad (5)$$

with β a learnable temperature. W_g is initialised to zero and b_g to -2, so the gate starts at $g \approx 0.12$, and every block is wrapped in LayerScale [36] with residual branches multiplied by a learnable per-channel gain initialised to 10^{-2}. In practice this means the network starts life as basically its own convolutional sub-network, the central-difference stem plus a plain classifier, and only gradually recruits attention, with the Born channel inside attention only earning gradient through its residual as training proceeds. This matters more than it might sound: in controlled probes at matched budget, removing LayerScale let the freshly initialised attention inject noise into the stem and slowed convergence relative to either component trained on its own. The converged gate value g is reported directly to the operator, which makes the mechanism itself auditable rather than a black box.

Measurement-entropy regularisation. Let $\mathscr{H} = \mathbb{E}_t[-\sum_j A_{ij}\log A_{ij}]$ be the mean row entropy of the attention matrix. Too low means attention has collapsed onto a single token and stopped integrating any real evidence; too high means

it has diffused into a uniform mixture and is carrying nothing at all. A parameter-free two-sided hinge keeps it inside a band:

$$\mathscr{L}_{\text{coh}} = [\lambda_{\text{lo}} \log T - \mathscr{H}]_+ + [\mathscr{H} - \lambda_{\text{hi}} \log T]_+, \tag{6}$$

with $(\lambda_{\text{lo}}, \lambda_{\text{hi}}) = (0.30, 0.80)$.

D. Quantum evidence basis

Let $\{|e_b\rangle\}_{b=1}^{B}$ be the first B computational basis states and $V(\phi)$ a second rotation circuit of the same family described above. The evidence basis is $|\phi_b\rangle = V(\phi)|e_b\rangle$, which stays exactly orthonormal for every value of ϕ. Measuring the output token states against it gives Born probabilities

$$\Pi_{tb} = |\langle \phi_b | \psi_t \rangle|^2, \qquad \sum_b \Pi_{tb} \le 1. \tag{7}$$

Π does two jobs at once, and that dual role is really the whole design point. Its token-wise average is concatenated straight into the classification feature vector, and the same tensor, not an approximation of it and not some separate gradient-based surrogate map, is exported unchanged as the explanation. A lightweight auxiliary classifier trained on $\overline{\Pi}$ with ordinary cross-entropy pushes each mode to become class-selective, so that after training every mode can be *named* after the family it responds to most. Those names are read off the trained model rather than assigned by hand.

E. Decoupled evidential head and vacuity

A Dirichlet distribution is completely described by a direction on the simplex and a concentration. The standard evidential formulation [21] predicts all K concentrations jointly; we predict the two factors separately, which turned out to matter quite a bit in practice. The direction (which class) and magnitude (how much evidence) are

$$\pi = \text{softmax}(W_c h), \tag{8}$$

$$E = \exp\big(\text{clip}(f_e(\text{sg}[z]))\big), \tag{9}$$

from which the Dirichlet concentrations, total evidence, and vacuity follow:

$$\alpha = \mathbf{1} + E\,\text{sg}[\pi], \qquad S = K + E, \qquad u = \frac{K}{S}, \tag{10}$$

where $\text{sg}[\cdot]$ denotes a stop-gradient. The subjective-logic reading stays intact: belief masses are $b_k = E\pi_k/S$, vacuity is $u = K/(K+E)$, and the projected probability under a uniform base rate is $\hat{p}_k = (1 + E\pi_k)/(K+E)$. We report π as the belief direction and u as the uncertainty, exactly the separation subjective logic draws between what I believe and how much I actually have to go on.

We verified three consequences of this decoupling empirically, in budget-matched probes. First, the class decision keeps a clean log-loss gradient, since $\arg\max_k \hat{p}_k = \arg\max_k \pi_k$, so training the class decision reduces to ordinary cross-entropy on $W_c h$. The joint parameterisation, by contrast, routes every gradient through α, whose Brier fit saturates early and whose usual KL regulariser drives the pre-activation negative until the softplus derivative collapses. We call this evidence starvation, and we measured it directly: swapping the joint head for the decoupled one recovered several points of eight-way accuracy at an identical training budget. Second, adding uncertainty cannot cost detection accuracy, because the magnitude branch only ever reads a stop-gradient copy of the shared representation. Letting its own objective (especially PUES, below) reshape those features would be circular, and we measured it to be costly when we tried it. Third, suppressing evidence becomes a strictly one-dimensional problem: because all of the concentration lives in the single scalar E, an out-of-taxonomy constraint cannot distort the class direction at all.

Training combines a cross-entropy term on the direction, a Brier-style fit on the concentration, and a term that ties the magnitude to the model's own correctness:

$$\mathscr{L}_{\text{ev}} = \text{CE}(\pi, y) + \tfrac{1}{2}\mathbb{E}\big[\|y - \hat{p}\|^2\big] + \lambda_c \,\text{BCE}\big(1 - u, \mathbb{1}[\hat{y} = y]\big), \tag{11}$$

with $\lambda_c = 0.4$. We use the expected sum-of-squares (Brier) form of the fit rather than the more standard digamma-based maximum likelihood, because the latter's gradient effectively vanishes while the Dirichlet is still near-uniform, which is exactly the regime a freshly initialised model starts in.

Two details here were forced on us by direct measurement rather than chosen for elegance, and we think they matter enough to spell out. We drop the usual KL regulariser entirely. In the joint parameterisation the KL term is what stops evidence from being placed on the wrong classes, but here the wrong-class evidence is already $E\pi_k$, which cross-entropy drives to zero on its own by sharpening π. The KL term therefore has nothing left to act on except the shared magnitude E, and leaving it in causes short-schedule models to saturate at $u \approx 1$ everywhere and the open-set channel to die. We also added the correctness term specifically because it is what makes vacuity useful *quickly*. The Brier fit does reward concentration, but only indirectly, so on a short schedule the magnitude head barely moves from its initialisation without it. Training $1 - u$ directly as "the probability this decision is correct" makes the channel usable within about two epochs. In a controlled leave-one-attack-out probe, adding this term moved the AUROC with which vacuity separates a held-out family from genuine samples from 25.6% (effectively inverted) to 73.9%.

F. Pseudo-Unseen Evidence Suppression

An intrinsic uncertainty score is known to be weaker than one trained against an actual proxy for "something else" [23], but for attack detection there is no natural pool of outliers lying around. We build one out of the taxonomy itself. Take two training images x_a, x_b from different attack families and mix them with $\lambda \sim \text{Beta}(2,2)$: $\tilde{x} = \lambda x_a + (1-\lambda) x_b$. This blend carries a superposition of two forensic signatures that no single physical or generative process could ever produce, so it is outside the taxonomy by construction. We constrain its evidence to sit

well below that of real samples:

$$\mathscr{L}_{\text{PUES}} = \frac{\left[\|\tilde{e}\|_1 - \rho\,\|e\|_1^{\text{sg}}\right]_+}{\|e\|_1^{\text{sg}} + 1}, \qquad \rho = 0.25. \tag{12}$$

The relative form here matters and was arrived at empirically. An absolute target ($\|\tilde{e}\|_1 \to 0$) is trivially satisfied by the degenerate solution of emitting zero evidence *everywhere*, collapsing the classifier to a uniform guess. We saw exactly that failure, 12.5% accuracy over eight classes, in an early version. Tying the constraint to the batch's own in-taxonomy evidence removes that shortcut entirely, because the fit term still demands sharply peaked posteriors on the real samples in the same batch. This is worth distinguishing from mixup [37], which asks for a mixed *label*. PUES asks for the *absence of belief*, which only makes sense because the head is evidential in the first place.

G. Evidence localisation

A lightweight feature-pyramid decoder fuses the 7^2 token map with the 14^2 and 28^2 intermediate stem maps and predicts M at half input resolution, supervised with a BCE plus soft-Dice objective against the operator's own ground-truth mask. Unlike depth-map supervision [1, 3], this target is the actual support of the forensic evidence, and it is well-defined for digital attacks just as much as for physical ones.

H. Risk calibration

Attack families are not equally harmful. With per-family severity weights ς and evidence area $a = \overline{M}$,

$$r = (1-u)\sum_k \hat{p}_k \varsigma_k + \lambda_u\, u + \lambda_a\, a. \tag{13}$$

The first term is the expected harm under the posterior, damped by how much the model actually knows. The second charges an explicit unknown-attack premium, so a confidently-unknown sample gets escalated rather than quietly waved through, which is the behaviour an operator needs and a plain softmax pipeline structurally cannot give. The third term rewards spatially corroborated evidence. All three coefficients, and the band edges that turn r into a Low/Medium/High/Critical label, are treated as deployment policy rather than learned parameters. No training signal touches them, deliberately, since risk is the one part of the pipeline an operator has to be able to audit and adjust, and it is the last place we want to hide a silently-fitted parameter. A vacuity of $u \geq 0.55$ overrides the verdict and refers the sample to a human reviewer.

I. Evidence-grounded reasoning

Every forward pass emits a typed evidence record: the posterior $\hat{p}$, the vacuity u, the risk score and band, the top facial regions of M and their mean response, the area and spatial dispersion of M, the three strongest quantum modes with their learned names and activations, the spectral descriptors, and the mean adaptive gate value. Language is generated only from this record, through either of two interchangeable back-ends. The grounded realiser, used for every result in this paper, is a deterministic generator where every sentence is emitted by a rule whose guard is a numeric condition on the record, so every sentence traces back to a measured quantity, the hallucination rate is zero by construction, and it costs under a millisecond with no network call at all. The constrained-MLLM back-end instead hands the record to a hosted multimodal language model with an instruction to paraphrase only, never to add anything, and then runs an automatic verification pass that re-checks every numeric claim in the generated text against the record and flags anything unsupported. This inverts the usual dependency in MLLM-based forensics [14, 15], where the language model sees the image but not the detector's internal state. Here the detector's state *is* the input, so the explanation cannot contradict the decision it is describing.

J. Training objective

$$\mathscr{L} = \mathscr{L}_{\text{ev}} + w_m\mathscr{L}_{\text{loc}} + w_b\mathscr{L}_{\text{basis}} + w_s\mathscr{L}_{\text{sup}} + w_p\mathscr{L}_{\text{PUES}} + w_c\mathscr{L}_{\text{coh}}, \tag{14}$$

with $(w_m, w_b, w_s, w_p, w_c) = (0.6, 0.25, 0.3, 0.15, 0.02)$ throughout. $\mathscr{L}_{\text{sup}}$ is a cross-entropy on the marginal over the three super-classes (genuine, physical, digital), obtained by summing $\hat{p}$ within each group. It imposes the taxonomy's own hierarchy, so a mix-up between two families inside the same super-class is penalised less harshly than one that crosses the physical/digital boundary.

V. Experiments

A. Setup

Every model in every table is trained by us, from scratch, under an identical budget: 10 epochs, batch size 48, AdamW ($\text{lr} = 2.5 \times 10^{-3}$, weight decay 0.02) with a one-cycle schedule, gradient clipping at 3.0, and 112×112 random crops from the 128 px store with horizontal flips and mild photometric jitter. We deliberately use no geometric warps, since those would destroy the periodic forensic signatures we are trying to detect in the first place. No model uses pretrained weights, so the comparison isolates architecture and objective rather than pretraining corpus. The operating threshold is fixed at the equal-error point of the validation split and applied unchanged to test. Training and inference both run on CPU only.

The baselines are ResNet-18 [38], a width-matched residual CNN and the classical binary detector; ViT-Tiny [39], a patch-16 transformer with ordinary attention, included as the plain-transformer control from the same architecture family as the ViT-B/16 entry that underperforms in [9]; and CDCN++ [2, 3], a central-difference trunk with pixel-wise map supervision. CDCN++ is the strongest lightweight texture detector available to us and the one that shares our stem, so comparing against it isolates exactly what the transformer, the Born channel, the evidential head and the reasoning stack add.

B. Seen-attack unified detection (P1)

Table I tells three stories at once. First, the protocol is nearly saturated for every model with a convolutional front end. ResNet-18, CDCN++ and X-UAD all sit around 99.9% AUC and under

TABLE I
Protocol P1 (seen attacks, identity-disjoint, ID-consistent).
Metrics follow ISO/IEC 30107-3 [35]. Best per column in bold.

Method	Params (M)	ACER ↓	AUC ↑	EER ↓	ECE ↓
ResNet-18 [38]	1.59	0.92	**99.92**	**0.49**	2.11
ViT-Tiny [39]	0.64	49.75	73.43	34.65	**0.67**
CDCN++ [2]	**0.53**	**0.71**	**99.92**	**0.49**	3.44
X-UAD (ours)	1.29	1.46	99.89	0.76	**1.94**

ViT-Tiny's low ECE is an artifact of a detector operating near chance; it is not a meaningful calibration result. Among the three working detectors, X-UAD is best calibrated.

1.5% ACER. This mirrors what the unified-detection literature reports on its own seen-attack protocol [9], and it is a large part of why we do not treat P1 as the interesting experiment in this paper. We did check whether the remaining gaps are real rather than noise instead of assuming it. Pairwise McNemar exact tests on discordant errors reject equality on the eight-way task throughout, reaching $p = 4.2 \times 10^{-22}$ against ResNet-18. Only the BPCER gap is small enough to be within sampling noise.

Second, the plain patch-16 transformer fails outright, at 49.8% ACER. With 16×16 patches it never gets to see the high-frequency texture that the physical attack families actually live in, which is the same weakness that puts ViT-B/16 behind ResNet-50 in [9], in a much starker form here. This is exactly why X-UAD puts a central-difference convolutional stem underneath its transformer instead of tokenising raw pixels directly.

Third, calibration is where the models genuinely separate. X-UAD's expected calibration error is 1.94% against 2.11% and 3.44% for the two convolutional baselines, and it gets there while also carrying a bounded uncertainty signal usable as an open-set score, which neither baseline has at all. That comes at a cost of 0.75 ACER points and about 1.6 points of eight-way accuracy against the strongest baseline. The ablations in Section V-G trace that cost specifically to the components that build the uncertainty channel, not to the attention mechanism itself.

C. Unseen attack families (P2)

This is the operationally decisive question. What happens when an attack family the model has genuinely never seen shows up at test time? Table II holds out one family at a time and reports four separate things: the ISO error on the full test set, the eight-way family accuracy, the detection rate of the held-out family at the unchanged threshold, and the one capability no baseline has at all, the AUROC with which the model's own uncertainty channel ranks the held-out family above genuine samples.

Every method's error rate jumps from around 1% to somewhere in the 29 to 41% range once a family is withheld, which confirms that this really is the regime the unified-detection literature has flagged as unsolved. Within that regime, X-UAD rejects the unseen family most often on average (77.5% against 74.7% and 70.2%), and its uncertainty is a purpose-built channel rather than an incidental by-product of a softmax. Its novelty AUROC clearly leads on two of the four folds, replay (71.0 against 49.3 and 35.3) and face-swap (87.9 against 81.7 and 74.4), and trails on the other two, landing roughly level with CDCN++ on the mean and well ahead of ResNet-18.

We report the "is it rejected" question and the "does uncertainty rank it above known attacks" question separately and deliberately, because they answer different questions for an operator. Whether the held-out family is rejected at all is the operational question; whether uncertainty ranks it above the seen attacks is the analyst's question, the one that actually flags that a new attack vector has appeared. The adversarial fold makes the distinction concrete. X-UAD rejects 98.2% of a family it never trained on while still assigning it *low* vacuity, because the perturbation sits on top of an attack whose underlying evidence the model does recognise. That fold is also where our uncertainty channel is weakest, at 43.0 AUROC, meaning the ranking is mildly inverted. We have no satisfying account of why, and we flag it as an open weakness rather than rounding it off.

D. Cross-domain transfer (P3)

This is the least flattering table in the paper, and we report it in full rather than trimming it down. Domain B differs from Domain A in colour, alignment, resolution and compression history all at once, and every single method collapses under the shift. With the threshold fixed on the source domain, BPCER runs high across the board, meaning the detectors reject almost every genuine face from the new domain outright. Ranked by threshold-free AUC, CDCN++ retains the most signal at 65.96, while ResNet-18 (46.78), ViT-Tiny (50.18) and X-UAD (50.90) all sit at or below chance.

We draw two separate conclusions from this rather than one combined one. The benign conclusion is that this protocol is simply harsher than the cross-dataset protocols usually reported elsewhere, because the shift here is joint across four factors rather than a single one. The uncomfortable conclusion is that none of the mechanisms in this paper, ours very much included, addresses capture-domain shift. That is a genuinely different problem from attack-family shift, and it would need the domain-generalisation machinery of [4, 16, 17] layered on top. X-UAD does not lead this table and we make no claim that it does. We think reporting that plainly is more useful to the field than quietly leaving it out.

E. Calibration and evidence localisation

The softmax baselines are overconfident in the classic, well-documented way [40]. Their confidence mass piles up near 1 regardless of whether the prediction is actually correct, which is exactly why their ECE is high and why their raw confidence score is close to useless as an abstention signal on its own. X-UAD's Dirichlet vacuity, by contrast, is bounded and bimodal, and as Table II shows, it reliably ranks unseen families above genuine samples on two of four folds.

Table IV reports intersection-over-union against the generator's own mask. We separate out the families whose evidence genuinely occupies a sub-region of the frame (3D mask, face-swap, reenactment, adversarial) from the ones that alter the whole capture (print, replay, synthetic), since the latter have near-constant masks and a trivially high IoU that would tell us nothing about localisation quality. The region-level mean is the number that actually measures whether the localiser works.

TABLE II

Protocol P2, leave-one-attack-out. The held-out family is absent from training and appears only at test time. Means are over all four folds. Best per column within each metric block in bold.

Method	Replay	3D mask	Face-swap	Adversarial	Mean
ACER (%), lower is better					
ResNet-18 [38]	51.96	30.38	49.96	30.71	40.75
CDCN++ [2]	**25.67**	39.92	**25.25**	**25.58**	**29.10**
X-UAD (ours)	26.79	**21.42**	38.33	39.67	31.55
Eight-way family accuracy ACC_8 (%), higher is better					
ResNet-18 [38]	53.94	62.66	51.39	61.53	57.38
CDCN++ [2]	60.62	65.07	73.23	**63.84**	**65.69**
X-UAD (ours)	**60.95**	**69.01**	**73.34**	57.59	65.22
Detection rate of the held-out family (%), higher is better					
ResNet-18 [38]	**100.00**	61.67	**37.17**	**99.92**	74.69
CDCN++ [2]	93.08	34.25	60.33	93.08	70.19
X-UAD (ours)	88.00	**88.08**	35.58	98.17	**77.46**
Novelty AUROC (%), higher is better					
ResNet-18 [38]	35.29	56.69	74.40	36.88	50.82
CDCN++ [2]	49.28	**66.57**	81.66	**55.08**	**63.15**
X-UAD (ours)	**71.03**	50.41	**87.86**	43.04	63.09

Novelty AUROC is the separation between the held-out family and genuine samples using the model's own uncertainty channel. Values below 50 indicate an inverted ranking, which we report rather than suppress.

TABLE III

Protocol P3, cross-domain transfer A → B, no adaptation. Every method degrades heavily. Best per column in bold.

Method	ACER↓	AUC↑	EER↓	ACC_8↑	ECE↓
ResNet-18 [38]	49.79	46.78	51.87	2.29	83.26
ViT-Tiny [39]	50.00	50.18	50.23	10.31	82.47
CDCN++ [2]	**45.79**	**65.96**	**37.93**	**18.67**	**60.09**
X-UAD (ours)	49.71	50.90	49.98	3.65	81.61

TABLE IV

Spoof-evidence localisation, IoU (%) against the generator's own mask, protocol P1. ResNet-18 and ViT-Tiny have no localisation output.

Method	3D mask	Face-swap	Reenact	Region-mean
CDCN++ [2]	79.5	76.8	64.2	73.5
X-UAD (ours)	**83.6**	**78.1**	**71.6**	**77.8**

One methodological point belongs here rather than buried in the appendix. Predicted maps are binarised at a fixed threshold of 0.5, the natural midpoint of a sigmoid trained with a BCE plus Dice objective. That value is a convention, not a fitted quantity. We never swept it, on validation or anywhere else, and the same fixed value is applied identically to CDCN++ and to X-UAD.

F. Is the explanation actually faithful?

An explanation is only worth reading if perturbing what it highlights actually changes the decision. Table V reports deletion and insertion AUC [26] for the model's own attack score under three intrinsic explanation channels plus a random control.

The deletion result is clean. All three channels push the score down faster than random removal does, with the quantum evidence magnitude strongest at −0.040 and the localisation map close behind at −0.031. This is the property that matters most for an audit, since it says the pixels these maps highlight are genuinely load-bearing for the decision.

TABLE V

Faithfulness of the three intrinsic explanation channels, measured against the model's own attack score. Δ columns are relative to the random control.

Explanation map	Del.↓	ΔDel.↓	Ins.↑	ΔIns.↑
X-UAD localisation map	0.915	−0.031	0.917	−0.029
AQA attention rollout	0.932	−0.015	0.921	−0.025
Quantum evidence magnitude	**0.906**	−0.040	0.899	−0.047
Random (control)	0.946	—	0.946	—

The insertion result is not clean, and we report it as it stands. None of the three channels beats the random control on insertion; all three sit between 0.025 and 0.047 below it. We suspect this is a property of the benchmark rather than of the maps, since three of the eight families alter the whole capture, so restoring a small highlighted sub-region rarely recovers the decision while restoring scattered random pixels samples the whole-frame evidence more evenly. That is a conjecture, and we have not tested it. The honest summary is that our explanation maps pass the deletion test convincingly and fail the insertion test, and that anyone relying on these maps should know both halves.

G. Ablations

Table VI removes one component at a time at a reduced training budget so the sweep stays affordable. These rows are comparable with each other and not with Table I, and we would ask readers not to cross-reference the two.

The central-difference stem is clearly load-bearing. Removing it costs more than every other component combined, dropping eight-way accuracy by 21 points and region-level IoU by 12.

TABLE VI
Component ablation at a reduced training budget. Deltas are relative to the full model in the first row. These rows are comparable with each other but not directly with Table I.

Variant	ACER ↓	AUC ↑	ACC_8 ↑	IoU_{loc} ↑
X-UAD (full, reduced budget)	38.00	87.57	69.62	64.62
w/o Adaptive Quantum Attention	32.75 (−5.25)	91.05 (+3.49)	67.45 (−2.18)	63.36 (−1.26)
w/o evidential head (softmax)	30.54 (−7.46)	91.06 (+3.49)	72.07 (+2.45)	65.02 (+0.41)
w/o spectral branch	31.13 (−6.87)	88.44 (+0.88)	70.67 (+1.04)	64.54 (−0.07)
w/o central-difference stem	49.08 (+11.08)	73.64 (−13.93)	48.62 (−21.00)	52.63 (−11.99)
w/o pseudo-unseen suppression	26.04 (−11.96)	91.08 (+3.51)	72.90 (+3.27)	64.23 (−0.39)
w/o localisation supervision	33.58 (−4.42)	89.17 (+1.61)	70.56 (+0.94)	34.34 (−30.27)

TABLE VII
Converged adaptive gate g, the fraction of the attention score taken from the Born channel, averaged over heads, layers, and test samples of each family. Higher means the interference channel is doing more of the work.

Attack family	Genuine	Print	Replay	3D mask	Face-swap	Reenact	Synthetic	Adversarial
Mean Born-channel gate g	0.309	**0.434**	0.231	0.287	0.267	0.267	0.284	0.394

TABLE VIII
Inference cost, CPU, single 112×112 crop. X-UAD is the only model here that also outputs an evidence map, named modes and a rationale.

Method	Params (M)	GMACs	Latency (ms)
ResNet-18 [38]	1.59	0.368	16.8
ViT-Tiny [39]	**0.64**	**0.031**	**2.9**
CDCN++ [2]	0.53	0.070	13.1
X-UAD (ours)	1.29	0.119	16.9

Localisation supervision is what actually gives the evidence map meaning; without it, region-level IoU falls by more than half even though raw detection accuracy barely moves, which is a fairly clean demonstration that a saliency map obtained for free from a detector's gradients is simply not the same thing as a supervised evidence map.

Two rows deserve specific comment, because they both trade closed-set accuracy for the uncertainty channel. Removing PUES actually *improves* reduced-budget ACER, while removing exactly the capability the term was built to provide. We isolate that trade on the fold where it actually matters, the face-swap leave-one-attack-out protocol, where PUES is worth 5.4 novelty AUROC points and 5.2 points of family accuracy, at a cost of 2.2 ACER points. Adaptive Quantum Attention shows the same pattern. At reduced budget it costs ACER but buys calibration and localisation, and specifically on the leave-one-attack-out fold it buys 1.9 AUC and 1.9 accuracy at equal novelty AUROC. We report both directions honestly rather than only the flattering one.

H. Efficiency

X-UAD adds its extra capability without a large parameter cost. The rotation-circuit parameterisation costs $L\lfloor d/2 \rfloor$ angles per circuit against the d^2 of a dense projection, and the deterministic reasoning module runs in under a millisecond and adds zero learned parameters. At 1.29M parameters, X-UAD sits between ViT-Tiny and ResNet-18 in size while being the only model in the comparison that outputs a localisation map, a set of named evidence modes and a written rationale. Table VIII gives the full picture.

Table VII shows where the Born channel earns its place. The converged gate opens furthest on print (0.434) and adversarial (0.394), and sits lowest on replay (0.231). We had expected replay to be the clearest interference family, given its moiré beat, so this is not the pattern we predicted, and we report the measurement rather than the prediction.

VI. The Research Gap, Revisited

We restate the gap from Section II as five concrete deficits, and pair each one with the mechanism that addresses it and the measurement that shows the effect.

Gap 1: detection is unified only in-distribution. The unified corpora [9, 11] show near-saturated seen-attack numbers next to 22 to 44% ACER once attack families are withheld. Nothing in those architectures represents "this belongs to no family I know", because a softmax has to put all of its mass somewhere. *Mechanism:* the Dirichlet evidential head makes total evidence, and therefore vacuity, an explicit output, and PUES trains that output against an outlier proxy built from the attack taxonomy itself rather than an unrelated image pool. *Evidence:* the unseen-family detection rate and novelty AUROC in Table II, neither of which a softmax baseline can even produce, and the w/o-PUES row of Table VI.

Gap 2: attention is linear, but much of the evidence is interference. Moiré beats, halftone screens, and upsampling checkerboards are periodic structures whose sign carries no information. A dot-product score is sign-sensitive and has to spend capacity learning that invariance from scratch. *Mechanism:* the Born channel scores token pairs by a squared overlap, which is sign-invariant by construction, and the input-conditioned gate recruits it only where it helps. *Evidence:* the w/o-AQA row of Table VI, and the per-family gate values in Table VII. We should be careful about what the gate table does and does not show. It confirms the channel is genuinely used, with a mean around 0.31 and a clear per-family structure, and the ablation confirms

it contributes. It does *not* confirm our original hypothesis that the gate would open furthest on the most periodic families. Print and adversarial open widest, while replay, the family with the most obvious moiré beat, opens least. The mechanism works; our story about *why* it works is not settled by this measurement.

Gap 3: pixel supervision targets a liveness proxy, not the evidence itself. Depth maps [1, 3] are undefined for a digitally manipulated but physically genuine capture, so a unified detector built on them inherits a supervision signal that is meaningless for half of its own label space. *Mechanism:* we supervise the actual injected support directly, which our benchmark can provide because it knows exactly where it wrote the evidence. *Evidence:* Table IV's region-level IoU on families for which a depth target would carry zero information.

Gap 4: MLLM explanations are post-hoc narration. FFAA [14] and M2F2-Det [15] give the language model the raw image, not the detector's internal state, so the narrative is unverifiable and can contradict the decision it is meant to explain. *Mechanism:* we invert the dependency. The forward pass emits a typed evidence record, and language is realised from that record alone, with the optional hosted back-end constrained and then re-verified against the same record. *Evidence:* the localisation channel of Table V is the one actually quoted in the generated text, and it is also the channel that clears the random control on deletion. The grounded realiser's hallucination rate is zero by construction.

Gap 5: a score is not a decision. An operator has to act on something, and a scalar in $[0, 1]$ with no calibrated meaning and no severity model does not tell them what to do. *Mechanism:* the risk calibrator converts posterior, vacuity and evidence area into a bounded risk score with an explicit unknown-attack premium, so that a confidently-unknown sample is escalated to human review instead of silently passing through. *Evidence:* the abstention behaviour induced by the $u \geq 0.55$ rule on the held-out families of Table II.

VII. Limitations

We would rather state these plainly than let a reader discover them.

Simulated evidence. Our operators are models of physical and generative processes, not the processes themselves. They reproduce the artifact families the forensic literature has identified as discriminative, including halftone, moiré, sub-pixel lattice, blending boundary, resolution mismatch, checkerboard, spectral peak and gradient perturbation, but the joint statistics of a real re-capture are richer than any operator we can write down. Absolute numbers in this paper should not be compared directly against numbers reported on captured corpora. The comparisons that are valid are the ones made *inside* this paper, where every method sees identical data.

No external validation. Every number reported here comes from our own simulated benchmark. We release adapters that map the identical taxonomy, protocols and model code onto seven public capture-based corpora, and those adapters are regression-tested against synthetic fixtures that replicate each corpus's on-disk layout. They have not been run on the real corpora. Nothing in this paper should be read as validation on captured data, and we would ask reviewers and readers to treat the released configurations as an invitation to replicate rather than as evidence that replication has already happened.

Grayscale source imagery. Domain A is grayscale, which removes colour cues equally for every method and makes the task harder rather than easier. It does mean the colour-domain components of our print and replay operators are inactive there. Domain B is what exercises them.

Still images only. Temporal evidence, meaning rPPG, blink dynamics, and the frame-to-frame refresh beat of a replay attack, is entirely out of scope. The architecture would extend fairly naturally by treating a short clip as a token sequence, and the Born channel seems well suited in principle to the temporal periodicity of a refresh beat, but we have not evaluated that extension.

Localisation threshold and sampling. Predicted evidence maps are binarised at a fixed 0.5. That is a convention rather than a policy, it was never fitted on validation or anywhere else, and it is applied identically across methods. Separately, localisation metrics are computed over the first 600 test samples in dataset order, and the pixel-level AUC over a 200-sample subset of those. On this benchmark the sample is balanced, because the eight families interleave every eight samples by construction, but on a corpus whose ordering groups samples by family that sampling would be biased. Anyone porting this evaluation to another corpus should stratify it.

Ground-truth mask decoding. For corpora that ship annotation videos rather than generator masks, our adapter binarises the decoded annotation at 0.15. That value affects no number in this paper, since every family in our own benchmark carries an exact operator mask, but it governs any future run on a real corpus and it has not been validated against real annotations. Relatedly, where a corpus ships no mask at all, our adapter substitutes a smoothed face-region ellipse as a proxy target, and that proxy is not currently excluded from IoU reporting. Neither condition is triggered by any result here, but both would need handling before an external localisation number could be trusted.

Compute scale. Every experiment here runs on CPU with a 1.3M-parameter model. The public challenge results [11] make clear that large pretrained multimodal backbones are currently the strongest known unified detectors. Our contributions are orthogonal to backbone scale, since AQA is a drop-in attention mechanism and the evidential head, PUES, the localiser and the reasoning module can all attach to any encoder. Combining them with a LoRA-adapted large vision-language backbone is a natural next step and a substantially larger undertaking than this paper.

White-box adaptive adversaries. Our adversarial class is a transfer attack, crafted against a surrogate that shares no weights with any evaluated model. A white-box adversary with actual gradient access to X-UAD itself, including through the vacuity channel, is a meaningfully stronger threat model and is not evaluated here.

VIII. Directions We Think Are Genuinely Open

A few of the questions below go beyond what this paper set out to do, and we name them explicitly rather than folding them into the limitations list, because we do not believe any existing work in this space has tackled them yet.

Attacking the uncertainty channel itself. Almost all open-set and evidential work, ours included, evaluates robustness of the *class decision* under adversarial pressure. Essentially nobody evaluates whether the *vacuity signal* itself can be attacked independently. Because our vacuity head reads a stop-gradient copy of the shared representation, an adversary with white-box access could plausibly search for a perturbation that lowers u, manufacturing false confidence, while barely moving the class logits at all, since the two objectives are explicitly decoupled during training. Whether that decoupling, which is exactly what makes the uncertainty channel free with respect to accuracy, also makes it a uniquely soft target is as far as we can tell an open question that no prior evidential deep learning paper in this space has measured.

PUES beyond pairwise blends. We only ever mix two attack families at a time. Whether three-way or higher-order mixtures, or mixtures weighted by how forensically *similar* two families already are rather than a flat $\mathrm{Beta}(2,2)$, produce a better outlier proxy is untested. There is a plausible argument either way. Closer families might produce a harder and more informative training signal for the vacuity boundary, or they might produce blends that are accidentally too close to a real attack and corrupt the class-direction gradient. We see no existing work, evidential or otherwise, that has studied how the *choice of mixing partners* inside a taxonomy-derived outlier proxy affects the resulting open-set boundary.

A learned, rather than fixed, risk policy. Section IV-G deliberately keeps the risk calibrator's coefficients unlearned, for auditability reasons we still believe in. But nothing stops a clearly-labelled meta-model from learning site-specific coefficients from a small amount of labelled operator feedback, such as analyst overrides of the band, while leaving the underlying detector and its explanation completely untouched. That would let a deployment site's risk tolerance adapt over time without ever touching the part of the system responsible for the actual forensic decision, a separation of concerns we have not seen made explicit in either the anti-spoofing or the forgery-detection literature.

Cross-domain shift and attack-family shift as a joint problem. Every domain-generalisation technique for face anti-spoofing we are aware of [4, 16, 17] targets capture-domain shift alone, using only *seen* attack families. Every open-set technique, including PUES, targets attack-family shift on a *single* capture domain. Nobody has yet studied what happens at the intersection, where an unseen attack family arrives through an unseen capture domain simultaneously. Our own Table III suggests the joint case may not simply be the product of the two individual failure rates. It could be substantially worse, or the two shifts could interfere destructively and compound only mildly. We think this is a real gap in the field rather than a mere extension of our own limitations, and closing it would need a benchmark explicitly designed to vary both axes independently.

Explanation faithfulness under distribution shift. Table V measures faithfulness only on in-distribution, seen-attack samples. Whether an intrinsic explanation stays faithful when the underlying detector is operating in the P2 or P3 regime, which is to say whether the localisation map still points at something real when the class decision itself is unreliable, is untested here and, to our knowledge, untested anywhere else in the explainable-forensics literature. A model that is uncertain but still produces a confident-looking localisation map would be a genuinely dangerous failure mode for exactly the auditability reasons this paper cares about, and we think it deserves its own dedicated study rather than a single additional column in a results table.

IX. Conclusion

X-UAD detects physical and digital face attacks with a single model, reports how much it does not know, shows where the evidence sits, and explains itself in language that can be audited sentence by sentence. Three ideas make this combination possible: a Born-rule attention channel that matches the interference character of the underlying evidence at a parameter cost of $O(dL)$; an orthonormal quantum evidence basis that serves simultaneously as a classification feature and as the explanation itself; and pseudo-unseen evidence suppression, which manufactures an out-of-taxonomy proxy directly from the attack taxonomy and turns Dirichlet vacuity into a usable open-set score.

The gains concentrate exactly where unified detection is still unsolved, on unseen attack families, rather than on the already-saturated seen-attack task. The costs are real and we have reported them: a 0.75 ACER point deficit against the strongest baseline on P1, a mildly inverted uncertainty ranking on the adversarial fold, an insertion-faithfulness result that does not clear its control, and a cross-domain protocol on which no method in this paper, ours included, performs acceptably. Every number here comes from a simulated benchmark, and validating these mechanisms on captured corpora is the necessary next step rather than an optional one.

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